OpsAgents: test summary

Advanced Data Science project · 20 October 2026 · branch orchestrator

158	0	20/20	64
automated tests passed	failed	policy tests	benchmark cases, real models

How we test

- **Automated tests (pytest), 170 tests:** run on a real PostgreSQL database and the real policy engine (OPA). AI models are replaced by test doubles, so the tests are fast and repeatable.
- **Policy tests (opa test), 20 tests:** the Guardrail's approval rules.
- **Benchmark (opsagents-bench), 64 cases:** realistic cases with known right answers, run against the real agents and the real AI models.

Results

AREA	TESTS	RESULT
Guardrail	48	48 passed
Code Reviewer	27	27 passed
Drift Detection	30	18 passed, 12 skipped (drift data package not yet in the repository)
Dependency Resolver	12	12 passed
Orchestrator, GitHub, API, platform	53	53 passed
Total (pytest)	**170**	**158 passed, 12 skipped, 0 failed**
Guardrail policy (OPA)	20	20 passed

What the tests prove

- **Guardrail:** risk is scored from evidence and can only go up; it fails safe when the policy engine or the AI judge is down; nobody can approve their own change; high-risk production changes need two people; the audit log cannot be edited without detection.
- **Code Reviewer:** finds ML mistakes, only applies safe fixes, never merges, never reads outside the repository, and ignores instructions hidden in the code it reviews.
- **Drift Detection:** the statistics (PSI, Kolmogorov–Smirnov, chi-square, Jensen–Shannon) are correct and data is stored safely in PostgreSQL.
- **Dependency Resolver:** finds cycles without false alarms, even with OpenTofu, the AI model and the database all unavailable.
- **Platform:** runs complete or wait for approval correctly; a GitHub failure raises one alert and one investigation; secrets are removed from logs.

Benchmark: real models vs. rules only

AGENT	WITH AI MODELS	RULES ONLY
Orchestrator routing	13/15	8/15
Guardrail	20/22 **(1 unsafe approval)**	17/22 (4 unsafe)
Diagnostic / RCA	6/8 correct top cause	4/8
Code Reviewer	6/6 found, 0 false alarms	0/6
Dependency Resolver	6/6	6/6
Merge Assistant	5/5	5/5

What the benchmark found (next fixes)

- A prompt injection in an incident report steered the router.
- A patch that disables the authentication check was approved automatically.
- A production model rollback labelled "low" needed 1 approver instead of 2.
- The RCA agent once copied an unrelated document instead of reading the log.

Conclusion

All tests that can run pass. The AI models add the most to routing, risk judgement and code review. The benchmark found four concrete weaknesses, all fixable.

Full detail: OpsAgents-Tests-Explained.docx (every test explained) and docs/test-results/ (screenshots of each run).